

6 CONCLUDING REMARKS

This paper introduces the AI-Generated Index of Occupational Exposure (GENOE), a novel measure of susceptibility to job replacement by artificial intelligence. Using synthetic AI surveys conducted with advanced language models, we estimate the likelihood of job replacement by AI over one, five, and 10-year horizons. Our findings reveal a significant increase in exposure over time, with the average likelihood of job replacement rising from 0.28 in the next year to 0.38 and 0.44 over the next five and 10 years, respectively, underscoring the accelerating impact of AI on the labor market.

The GENOE methodology of synthetic AI surveys leverages the extensive knowledge and analytical capabilities of AI models to provide expert-like assessments. This approach offers a comprehensive view of occupational exposure by considering not only the tasks performed within each occupation but also ethical and social contexts that influence the likelihood of AI-driven automation. By doing so, it overcomes the limitations of previous measures by incorporating a more holistic task-based analysis and integrating other occupation-specific, social, and ethical considerations more organically and with fewer assumptions. This contrasts with prior methods that isolated tasks or abilities separately or relied on more predefined criteria for measuring exposure.

We validated the consistency of the index by varying key elements of the methodology and the prompts, such as calculating the task-by-task exposure, occupation name omission, and explicitly nudging the model to consider ethical and social resistance to automation considerations. Additionally, follow-up surveys with AI models uncovered key patterns including understanding and projections of current and expected technological capabilities, the routine versus non-routine nature of tasks, ethical considerations, regulatory environments, and cost-benefit analyses.

The GENOE index offers significant and practical uses and implications for academia, policymakers, industries, and individual workers. By identifying which types of tasks, occupations, and industries are most susceptible to AI-driven automation, the GENOE index can inform strategic decisions and policy formulations. For academia, it offers a robust framework for future research on labor market dynamics and technology's impact, opening new avenues for interdisciplinary studies on AI and employment.

Policymakers can use the index to develop targeted interventions and support measures for workers in highly exposed occupations, ensuring a smoother transition in the face of technological change. This data-driven approach can inform education and training policies, unemployment insurance programs, and economic development strategies. Industries can leverage the index to guide strategic decision on workforce development, technology integration, and long-term business planning. It can help companies anticipate skill gaps, plan for reskilling initiatives, and optimize human-AI collaboration.

For individual workers, the index provides valuable insights for career planning and skill development, helping them prepare for potential shifts in their occupational landscape. In the

broader societal context, the GENOE index contributes to the ongoing dialogue about the future of work, AI ethics, and the need for responsible AI development and deployment.

While the GENOE index represents a significant advancement in understanding AI's impact on employment, it is important to acknowledge its limitations. The index is based on current AI capabilities and projected advancements, and unforeseen technological breakthroughs or obstacles could alter these projections. Additionally, the index does not account for potential new job creation resulting from AI advancements.

Future research could expand on this work by incorporating more dynamic elements, such as the potential for job creation and transformation due to AI, and by exploring the interaction between AI exposure and other economic and social factors. Regular updates to the index will be crucial to reflect the rapidly evolving AI landscape.

In conclusion, the GENOE index provides a valuable tool for understanding and preparing for the impact of AI on the workforce. By offering a holistic, forward-looking perspective on occupational exposure to AI, it equips stakeholders across society with the insights needed to navigate the challenges and opportunities of the AI-driven future of work.

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